

Christopher Gilbert

MIT undergraduate pursuing Computer Engineering

Class of '29, Course 6-5 (Electrical Engineering with Computing), 5.0 GPA

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Experience

- **Wolfram Institute** | Summer Researcher | Independently designed algorithms for cellular automata structure-finding and met regularly with Stephen Wolfram to review my results and discuss direction | [Presentation](#) | 2026
- **Undergraduate Research** | In the Ultracold Quantum Gases Group (Prof. Martin Zwierlein): designing analog laser control circuit boards in KiCad, writing software for high-speed voltage-logging and plotting in C, as well as PCB assembly/soldering and sheet metal fabrication | IAP '26–Spring '26
- **Wolfram Summer Research Program** | *Novel Minimal Universal Classical and Quantum Gates*, a paper on finding a new universal gate set for quantum computers | 2024 | [Project](#) | [Paper](#)
- **Wolfram Emerging Leaders Program** | Year-long intense mathematical and computational research on a student team studying functional roots | 2024–2025 | [Project](#)
- **World Science Scholar** | Completed two years of challenging coursework in topics such as general relativity, computational analysis, and evolutionary medicine | 2021–2024
- **MIT Beaverworks Summer Institute** | Quantum Software | 2023 | [Presentation](#)

Projects

- **Minecraft Computer** | Turing-complete computer. Built with very little prior knowledge of computer architecture, and developing custom RAM, ALU, and ISA over several years | [Youtube Video](#)
- **GPU Structure Finder** | While working for Wolfram Institute, developed an OpenCL program employing bitpacking and boolean arithmetic to search over 100 trillion cellular automata initial conditions | [Presentation](#)
- **Exact Boolean Synthesis** | Rust implementation of an algorithm to find gate-minimal circuits for up to 5-variable boolean functions by means of exhaustive search over k-gate boolean circuits | [Project](#)
- **Large-Scale Gravity Sim** | A many-particle gravitational simulation for visualizing chaotic systems written in Python | [Project](#)
- **3D Renderer** | A software renderer for wireframe, raster, and ray-traced scenes written in Rust | [Project](#)
- **Combat Robotics Club** | Developed an antweight ring-spinner shuffler for the Fall 2025 competition | 2025

Skills

- **Current Classwork** | Computation Structures (6.191), Introduction to Algorithms, Algebra I

- **Previous Classwork** | Differential Equations, Linear Algebra, Multi-variable Calculus, C and Assembly Programming, Discrete Mathematics, Physics Mechanics and Electromagnetism
- **Programming** | Fluent with C, Rust, Mathematica, and Python, and have completed numerous projects in each
- **Fabrication** | Familiarity with CAD for 3D printing, sheet metal, PCB design, and other tools and techniques for design and construction
- **Software** | Linux, Git, KiCad, LTspice, Onshape, Autodesk Inventor

Awards and Honors

- **Davidson Fellows Scholarship** | For *Novel Minimal Universal Classical and Quantum Gates* | 2025
- **Presidential Scholars Winner** | 2025
- **National Merit Scholarship Winner** | 2025
- **Rise for the World** | Global Finalist | for Creating STEM Education Videos | 2023
- **Award-Winning Author** | *Quantum Mechanics for Kids: a Humorous, Easy-to-Read, Math-Free Book on a Very Perplexing Subject* | 2019 | Nonfiction Authors Association Silver Medal | [Goodreads](#)
- **XVII Golden Key International Piano Composition Competition** | “For Lewiston” | 2023
- **Sempre Musick Winter Serenade Competition** | 1st Place | 2024 | [Organ Performance at Methuen Memorial Hall](#)

Service and Extracurriculars

- **Mentor and Presenter** | Junior World Science Scholars | 2023–2024
- **Author Talks and [YouTube Channel](#)** | On quantum mechanics, calculus, artificial intelligence, and more | 2019–2024
- **Guest Organist** | Basilica of Saints Peter and Paul, Lewiston, Maine | 2023–2026 | [Recording](#)
- **Psalterium Institute of Maine Chamber Choir** | 2023–2025
- **Basilica of Saints Peter and Paul Choir** | 2023–2026